

JORDAN NEWMAN

Systems Software Engineer | Kernel, Networking, Embedded Firmware

Marlboro, NJ | 347-739-4731 | phoem@mac.com | Private GitHub repository access available upon request

SUMMARY

Systems software engineer with deep C, FreeBSD, Linux, networking, security, kernel, and embedded firmware experience. Built production software from kernel modules and packet analyzers to high-performance web servers and embedded AVR firmware. Operated infrastructure at ISP/CDN scale across 3,000-4,000 servers, 10 CDN locations, and 65+ Gbps peak traffic.

TECHNICAL SKILLS

- **Languages:** C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic
- **Systems:** FreeBSD, Linux, kernel modules, system calls, bootloaders, operating systems, kqueue, GDB, TCP/IP
- **Networking/Security:** DDoS detection/mitigation, packet capture, DNS analysis, libpcap, IPFW, authentication, telemetry
- **Embedded:** Embedded C, Atmel AVR, SPI, UART, I2C, Arduino, circuit design, soldering, hardware troubleshooting
- **Cloud/DevOps:** Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, security scanning, deployment automation
- **Databases/Web:** MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

SELECTED TECHNICAL HIGHLIGHTS

- Built and operated infrastructure spanning 3,000-4,000 servers, 10 CDN locations, and 65+ Gbps peak traffic.
- Developed PrimeHTTPD in C around FreeBSD kqueue; supported 150,000+ concurrent connections across ~200 production servers.
- Created FreeBSD kernel modules for security, filesystem virtualization, and filesystem observability in customer-facing production systems.
- Developed libpcap-based network and DNS analyzers for real-time traffic visibility, DDoS analysis, and DNS recursion attack detection.
- Developed embedded C firmware for battery-powered wireless Atmel AVR smart smoke/CO2 detector prototypes using SPI, UART, and I2C.
- Built custom x86 OS kernel and bootloader with protected mode, IDT, heap allocation, port I/O, VGA output, and GDB debugging.

SELECTED SYSTEMS PROJECTS

- **TAFOS Operating System:** Developed educational x86 OS kernel in C and assembly with custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.
- **KeepClean:** Developed FreeBSD kernel module hooking syscalls to monitor execution activity, protect system assets, and report telemetry to userspace services.
- **VirtualDir:** Developed FreeBSD kernel module remapping filesystem paths through syscall interception; deployed customer-facing across approximately 30 production servers over 5-8 years.
- **StatCache:** Developed FreeBSD kernel module instrumenting stat() and lstat() activity for filesystem visibility and performance analysis.
- **PrimeDump:** Developed libpcap-based traffic analyzer decoding Ethernet, IP, TCP, UDP, and ICMPv6 headers with ncurses real-time views and IPFW integration.
- **PrimeDNSTop:** Developed DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source/domain activity.
- **AVR Smart Smoke/CO2 Detector:** Developed contract firmware in embedded C for battery-powered wireless Atmel AVR prototypes using SPI, UART, and I2C interfaces.

PROFESSIONAL EXPERIENCE

Advantive - Senior Development Specialist

Remote / Tampa Bay, FL | July 2022 - Present

- Implemented hands-on Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows across engineering teams.
- Partnered with architecture teams to improve deployment reliability, DevOps policies, and production software delivery practices.

DDI Systems - Senior Development Specialist

Manalapan, NJ | November 2021 - July 2022

- Modernized legacy VB.NET software by porting components to C#, replacing deprecated and insecure implementations.

ISPrime LLC - CEO

Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization, operational cleanup, and software platform improvements for production hosting infrastructure.

DDI Systems - Computer Programmer

Manalapan, NJ | April 2017 - April 2018

- Integrated ERP software with SellerCloud and Essendant systems, automating order, catalog, image, and SDS data workflows.

MFCXY, Inc. - Computer Programmer

Chicago, IL | December 2014 - September 2015

- Developed backend server features and Windows client integrations while identifying performance and security issues.

ISPrime Inc. - CIO / Partner

Weehawken, NJ | March 2001 - December 2014

- Built and operated infrastructure spanning 3,000-4,000 servers across 10 CDN locations and multiple datacenters.
- Sustained peak traffic exceeding 65 Gbps while supporting mission-critical customer workloads.
- Developed PrimeHTTPD, a high-performance C web server optimized around FreeBSD kqueue that supported 150,000+ concurrent connections and demonstrated performance competitive with, and in some workloads exceeding, nginx.
- Deployed PrimeHTTPD across approximately 200 production servers supporting high-traffic customer environments.
- Developed DDoS detection and mitigation systems protecting infrastructure during attacks involving up to 750,000 compromised hosts.

- Created PrimeDump and PrimeDNSTop, libpcap-based network analysis tools for traffic anomalies and DNS attacks.
- Developed production FreeBSD kernel modules including KeepClean, VirtualDir, and StatCache for security and filesystem observability.
- Built monitoring, telemetry, authentication, and security systems used across thousands of production servers.

FatWallet.com - Systems Administrator

Wisconsin / Remote | December 2000 - June 2001

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime.

EDUCATION

Brookdale Community College - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.